Spatial Inequities in Residential Energy Burden and HEAP Service Accessibility

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Academic Project Report

1. Problem Statement

In recent years, economic pressures, rising energy costs and the energy crisis have drawn growing public attention to household energy affordability. Defined as the proportion of income spent on household energy, the Residential Energy Burden (REB) has become a central metric in discussions of energy justice and urban inequality. According to the American Council for an Energy-Efficient Economy (ACEEE), "Low-income households spend three times more of their income on energy costs than higher-income households, often exceeding 10 percent of annual income" (Drehobl & Ross, 2016)[1]. Similarly, the U.S. Department of Energy's Low-Income Energy Affordability Data (LEAD) platform highlights pronounced geographic disparities across the southern and western United States[2], reframing energy affordability as both a socioeconomic and spatial-equity issue.

Los Angeles County, the most populous county in the United States, shows these disparities clearly. Home to 9.76 million residents and spanning 10,513 km², the region's low-density urban form, aging housing stock, and persistent heat conditions all contribute to uneven residential energy demand. As stated by the Neighborhood Data for Social Change (NDSC, 2023), "Energy burden varies dramatically across Los Angeles County, reflecting both housing efficiency differences and broader socioeconomic divides."[3] Understanding where high-burden communities are spatially concentrated is vital for improving energy equity and assistance to those in need.

This project analyzes the spatial clustering of REB within Los Angeles County and explores access to assistance locations such as LIHEAP (Low Income Home Energy Assistance Program) and HEAP (Home Energy Assistance Program) application centers, to help local government and residents address energy burden more effectively.

2. Approach

This project applies GIS spatial analysis to identify statistically significant REB hotspots and coldspots across Los Angeles County using Hot Spot Analysis (Getis-Ord Gi*) and evaluates whether REB hotspots have poorer access to energy-assistance service centers using the Near (Proximity) tool.

All datasets are standardized at the census-tract level, cleaned in Python and integrated within a GIS environment using ArcGIS Pro. By mapping spatial patterns of energy burden and measuring accessibility gaps, this project provides spatial evidence on these inequities to support more even planning of energy assistance.

- **Part I: Data Collection & Preparation**

- Search for and download all datasets needed for this project:
  - ✧ 2023 TIGER/Line census tract boundaries (statewide)
  - ✧ NDSC residential energy burden indicators (REB)
  - ✧ ACS socioeconomic data:
    - median income
    - tenure (renter occupancy)
    - housing age (year built)
  - ✧ HEAP/LIHEAP assistance center locations
- All datasets were imported into the ArcGIS Pro project geodatabase.
- To ensure spatial consistency, all datasets were standardized to the census-tract level and reprojected into NAD 1983 (2011) StatePlane California V.
- Attribute tables were checked to confirm that each dataset contained the full 20-digit GEOID required for accurate joining.

➢ Part II: Data Processing & Integration

- All tables except California Census Tract were imported into the geodatabase and joined to the Los Angeles County census tract layer by the key GEOID.
- The joined table was exported and cleaned in Python by removing invalid values and ACS error codes, converting incorrectly formatted numeric fields, handling missing or null values and computing derived variables such as Renter Percentage and Old Housing Percentage.
- The result is a complete, tract-level dataset suitable for spatial statistical analysis, which was re-imported into the geodatabase for further analysis.

➢ Part III: Spatial Analysis

- REB Hot Spot Analysis (Getis-Ord Gi*)
  - Hot Spot Analysis identifies statistically significant clustering patterns of REB across Los Angeles County.
    Using REB as the input variable, Gi* calculates z-scores, p-values, and Gi-bin classifications to distinguish:
    - high-burden hotspots
    - non-significant locations
    - low-burden coldspots
  - The output is a new spatial layer revealing where energy burdens are geographically concentrated, providing a statistical foundation for identifying vulnerable neighborhoods.

- HEAP Location Accessibility (Near) Analysis
    - To assess whether REB hotspots experience reduced access to assistance resources, the Near tool was used to compute the shortest Euclidean distance from each tract centroid to the nearest HEAP/LIHEAP service location.
    - The resulting NEAR_DIST metric quantifies accessibility at the tract level. By comparing these values between hotspot tracts and other tracts, the analysis evaluates whether spatial inequities in energy burden are amplified by inequitable access to assistance.

➢ Part IV: Visualization & Interpretation

- The results of the two analyses were mapped separately: LA County REB Hotspots Analysis and LA County HEAP Location Near Analysis.
- Symbology was adjusted on both maps for clarity.
- A layout was created for each map.
- The results of the hotspot and near analyses are interpreted below.

3. Maps

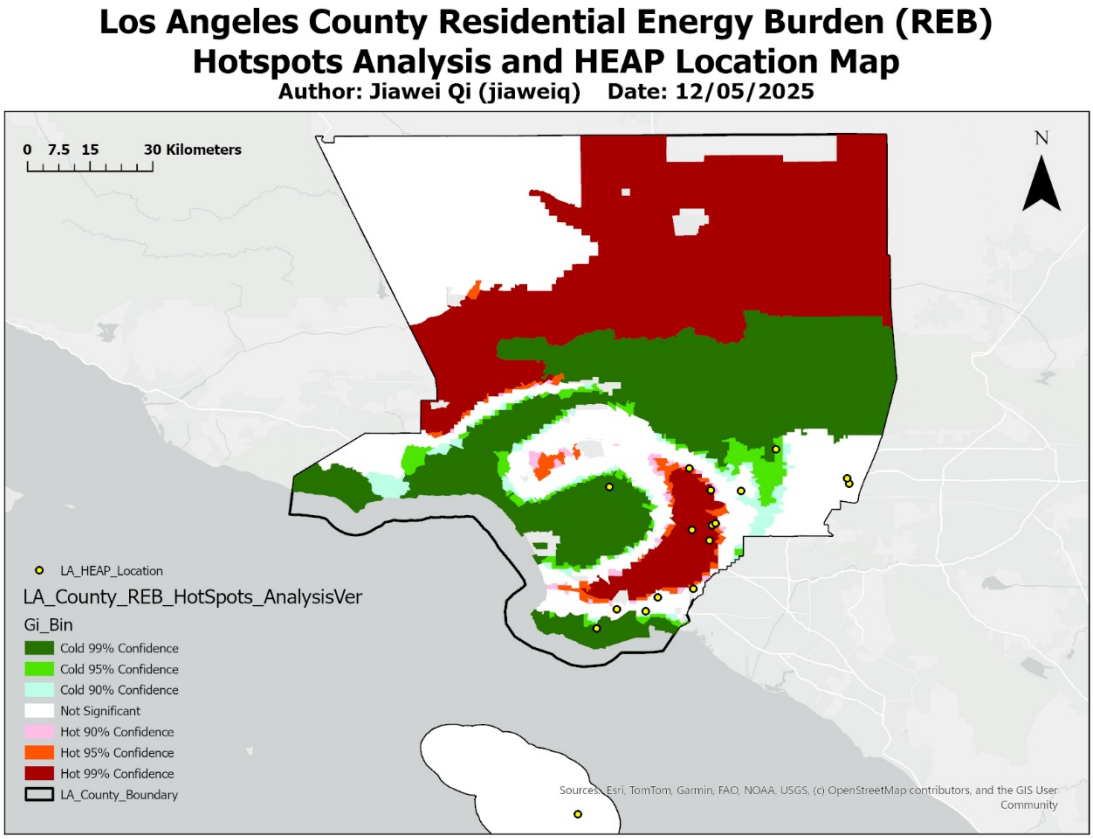


Figure 1 LA County REB Hotspots Analysis and HEAP Location Map Layout

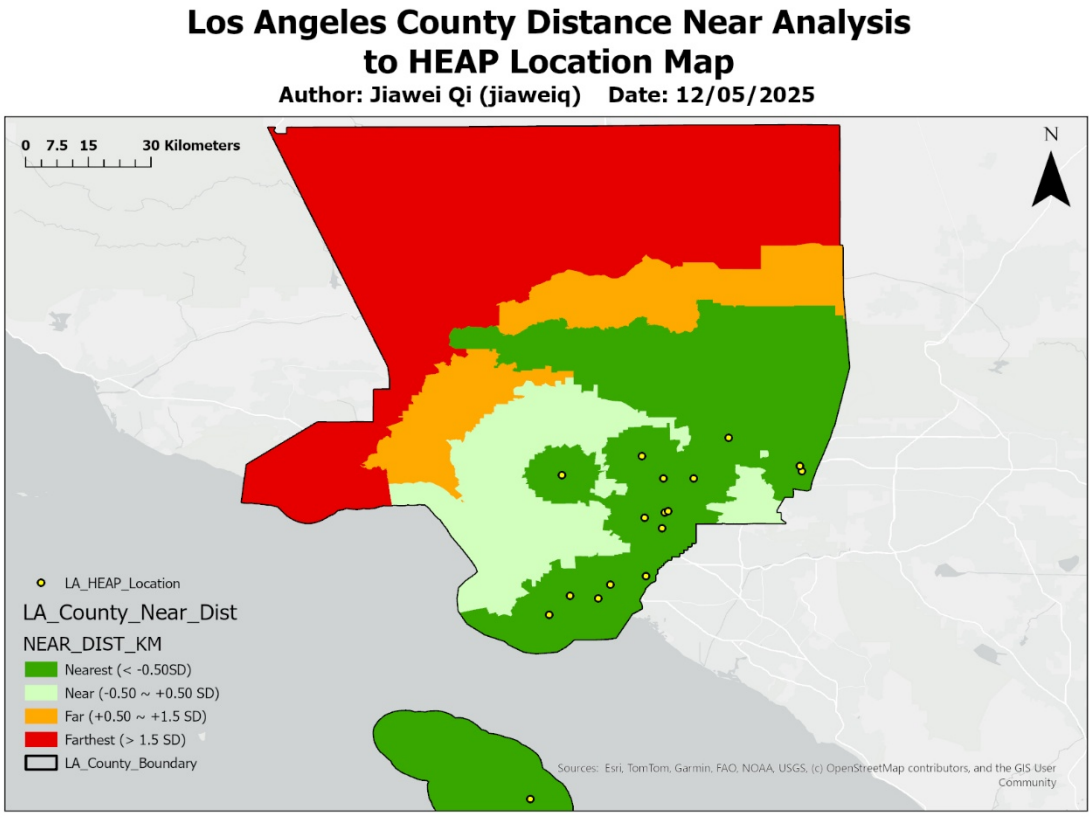


Figure 2 LA County Distance Near Analysis to HEAP Location Map Layout

4. Findings

1) Spatial Patterns of Residential Energy Burden (REB) Hotspots

The REB Hotspots Analysis map reveals a clear and non-random spatial clustering of residential energy burden (REB) across Los Angeles County. Hotspots, indicating significantly higher energy burden, are concentrated in the Antelope Valley, south of the Santa Clarita Valley, and Central Los Angeles regions. These areas are spatially contiguous and form a crescent-shaped pattern extending inland from central Los Angeles. Most of these places carry a severe energy burden, consistent with lower population density, aging housing, lower owner-occupancy rates, fewer high-value-added jobs in rural areas, and lower incomes.

Coldspots, indicating significantly lower energy burden, are found primarily in the central north part of the county and coastal areas, including Santa Clarita, Malibu, Beverly Hills, and parts of the suburban San Gabriel foothills. These areas align geographically with higher median incomes, more energy-efficient housing stock, and higher rates of owner-occupied units.

Approximately 30% of tracts show no significant clustering, including the northwest Santa Clarita Valley, the southeast San Gabriel Valley, Santa Catalina Island, and the transition areas between high- and low-burden clusters around downtown. These places may still have high energy demand, but higher incomes or more energy-efficient housing keep the burden on residents moderate.

In conclusion, the hotspot analysis clarifies that energy burden in Los Angeles County is not geographically uniform, but follows a core-to-periphery inequality model, where central south urban and northeast suburban tracts show significantly elevated energy hardship compared to other regions.

2) Statistical Validation of Hotspot Analysis (Histogram)

Beyond the map patterns, the faceted histogram of energy_burden_percentage by Gi_Bin provides statistical evidence that the detected hotspots and coldspots represent fundamentally different energy-burden regimes, rather than random spatial noise. Each small panel in the histogram corresponds to one Gi_Bin class (from –3 coldspot to +3 hotspot) and shows the internal distribution of REB values within that group.

The hotspot classes (Gi_Bin +1, +2, +3) shift the entire distribution to higher values and a wider spread. The +2 and +3 panels show most tracts with noticeably higher energy burden (many around and above 3–4%), and a few extreme outliers reaching very high percentages. This pattern is consistent

with the Antelope Valley, South LA, and central urban tracts identified in the map as high-burden areas characterized by older housing, lower incomes, and more energy-inefficient building stock.

The non-significant class (Gi_Bin 0) exhibits the widest and most heterogeneous distribution, with values spanning from low to moderate

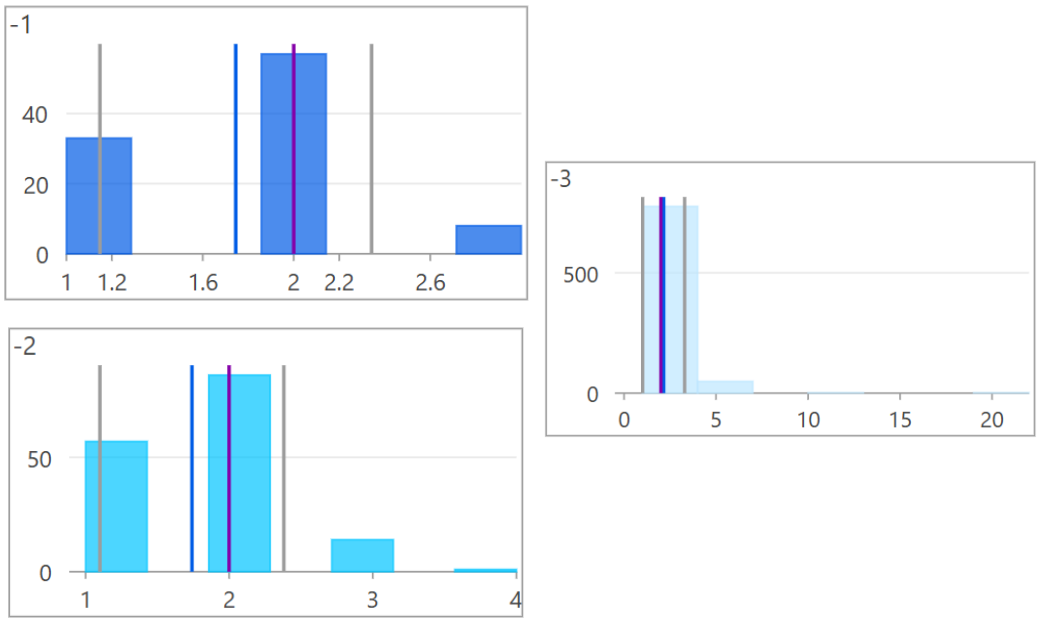
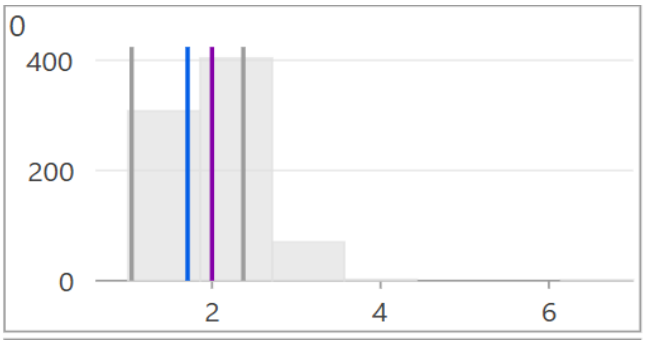


Figure 3 Histogram for Hotspots

energy burden. This matches the “transition zones” on the map—areas around downtown and the San Gabriel Valley that neither form strong high-burden clusters nor strong low-burden clusters.



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For the coldspot classes (Gi_Bin –3, –2, –1), energy burden values are tightly clustered in a relatively narrow low range (roughly around 1–2.5%). The bars

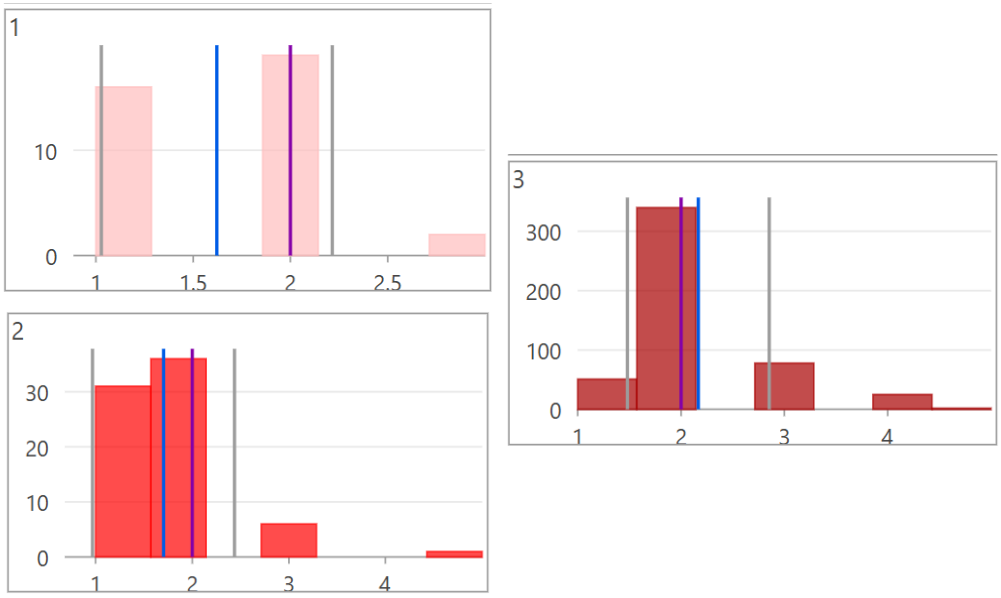


Figure 5 Histogram for Coldspots

are concentrated, and the spread is small, indicating that these tracts not only have low average energy burden but are also internally consistent. This confirms what is visually apparent on the map: coastal areas and high-income communities (e.g., Santa Clarita, Malibu, Beverly Hills and parts of the San Gabriel foothills) form stable low-burden clusters.

In conclusion, as Gi* significance moves from coldspots (–3) to hotspots (+3), both the typical level and the spread of energy burden increase. This statistical behavior reinforces earlier spatial findings: REB hotspots in Los Angeles County are genuinely different from the rest of the county in terms of their underlying energy cost burden, not just in terms of where they are located. It also supports the interpretation that the core-to-periphery inequality pattern is structurally embedded in the data, and not an artifact of the mapping choices.

3) Accessibility of Energy Assistance Resources (NEAR Analysis)

The NEAR distance classification map shows substantial variation in residential proximity to LIHEAP/HEAP service facilities. The majority of HEAP support locations are clustered within the Los Angeles Basin, especially around major population centers such as downtown, Pasadena, Long Beach and Compton.

Census tracts in these service areas fall within the Nearest distance category (< −0.50 Standard Deviation), meaning that large segments of densely

populated urban communities reside relatively close to the assistance centers.

However, tracts located in the northern, western, rural, and mountainous regions of the county show the greatest distances, falling within the (+1.5 Standard Deviation) classification, which represents the lowest service accessibility. These regions correspond to large-area tracts in the Antelope Valley, Palmdale, and Lancaster, where physical travel distance may pose a barrier, especially for low-income households with limited transit access.

4) Relationship Between REB Hotspots and Service Accessibility

Comparing the two maps reveals a partial misalignment between areas of highest energy burden and areas of highest service accessibility.

Many REB hotspots in South and central LA areas are located relatively close to HEAP service centers, so residents there can reach assistance relatively easily. Most of the moderate areas around HEAP centers show no significant energy burden.

In contrast, certain high energy burden zones in the western and northern portion of the county show moderate to far accessibility classifications. While most of the west is affluent with low or no significant energy burden, the north lags behind all other regions, with high energy burden and long distances to HEAP assistance centers.

The remaining regions are at moderate distances from these centers and carry moderate energy burdens.

These comparisons suggest that energy burden is more strongly associated with socioeconomic and housing characteristics than physical proximity to HEAP service locations. However, accessibility gaps may still disproportionately affect households with mobility constraints, especially in communities where car ownership is low and public transit is limited.

5) Implications for Energy Equity and Service Planning

Overall, energy burden in Los Angeles County is unevenly distributed, and energy assistance centers are unevenly located. Thus, the results support the policy argument that Los Angeles County may benefit from:

- Building more HEAP service centers or mobile intake units in the eastern basin and northern service deserts of Los Angeles County.
- Offering digital application methods or hybrid appointment systems to reduce physical access dependency.

- Continuing to retrofit residential housing for energy efficiency across all regions.
- Improving incomes, civil infrastructure and energy systems for all.

5. Future Work

1) Measure more contributors to REB

This project considers three contributors to REB: median income, housing age and tenure. Many other factors also affect it, such as education level, household composition and local policy, and including them would give a more complete picture.

2) Quantify contributors of REB

Beyond knowing which factors affect REB, the next step is to quantify how much each one contributes, rank them, and base recommendations on that ranking.

3) Integrate the two analyses

Combining the hotspot and near analysis results in a single model, possibly with additional GIS analysis tools, would test the relationship between burden and accessibility directly and make the findings more persuasive.

6. Data Sources

- **All Years Energy Burden Data (Los Angeles County)**

  Source: USC Lusk - Neighborhood Data for Social Change (NDSC, 2023)
  Description: Tract-level residential energy burden.
  Access URL: https://la.myneighborhooddata.org/2023/03/energy-burden

- **Year Built (B25034, 2023: ACS 5-Year Estimates Detailed)**

  Source: U.S. Census Bureau, American Community Survey (ACS)
  Description: Tract-level distribution of housing units by year built.
  Access URL: https://data.census.gov/table?q=B25034

- **Median Income (B19013, 2023: ACS 5-Year Estimates Detailed)**

  Source: U.S. Census Bureau, American Community Survey (ACS)
  Description: Tract-level median income per household.
  Access URL: https://data.census.gov/table?q=B19013

- **Tenure (B25003, 2023: ACS 5-Year Estimates Detailed)**

  Source: U.S. Census Bureau, American Community Survey (ACS)
  Description: Tract-level ratio of owner- to renter-occupied housing units.
  Access URL: https://data.census.gov/table?q=B25003

- **Census Tract Boundary (2020 Base, 2023 Version)**

Source: U.S. Census Bureau, TIGER/Line Shapefiles

Description: 2020 Census Tract polygons provide the geographic base.

Access URL: https://www2.census.gov/geo/tiger/TIGER2023/TRACT/

- **HEAP/LIHEAP Assistance Center Locations**

Source: Program listings from the Long Beach Community Action Partnership and the Pacific Asian Consortium in Employment, geocoded in ArcGIS Pro

Description: Point locations of HEAP/LIHEAP application centers in Los Angeles County.

7. References

[1] Drehobl, A., & Ross, L. (2016). Lifting the high energy burden in America's largest cities. American Council for an Energy-Efficient Economy. https://www.aceee.org/research-report/u1602

[2] U.S. Department of Energy. (2023). Low-Income Energy Affordability Data (LEAD) Tool. https://www.energy.gov/scep/lead-tool

[3] Neighborhood Data for Social Change. (2023). Energy burden and weatherization across Los Angeles County. https://la.myneighborhooddata.org